Motivation

- It's an honour to open this topic's session: Applied SE.
- hen talking about requirements, we should also define the requirements of the problem we're trying to solve.
- Storytime

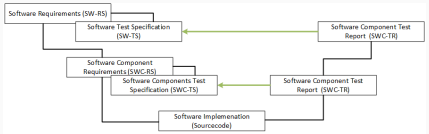
Motivation

- Fully automated traceability matrix
- Consistent and up-to-date documents from the source
- Store (requirements') meta-information with code

- It's fine manually. Requirements change find minimal delta-test? Change more than once. Hence we want this fully automated.
- Review multiple documents for a different contractor. No consistent set of documents were provided
- Jon Holt's term in introduction to MBSE: Documents are a live *view* of the system (not pretty pictures). CI/CD for software, why not for documents?
- It came with one document that contained all requirements and use-cases. It appeared to have been exported from some web-based tool. All the information therein is lost
- Also: Too technical, interface with management/business side
- 4'

Requirements and Traceability

- Requirements across system hierarchies
  - Implies the need for traceability
- *Traceability* from and to specification items
- Directions
  - Forward (*Impact*) from requirements specification to dependant documents
  - Backwards from verification artefacts to specification



Here Traceability only from items/issues

Proposed Solution

- *Requirement A-1.0*
  - Red button to shut down system
- *Implementation a* says implemented *A-1.0*
  - *A-1.0* (red button) changes to *A-2.0* (green button)
  - Use hashes instead of semantic versioning
  - Calculated automatically

DON'T PANIC

Why hashes: no tool or manual changes required, it's all derived

Let's see how it looks on an example

Example Requirement

```
Name: VCD Writer Inputs
Topic: ReqsDocument
Description: The output from ...
Rationale: Make the process as ...
Status: external
Owner: development
Effort estimation: 1
Invented on: 2020-05-30
Invented by: default
Type: requirement
```

Hash is calculated over *Name*, *Description* and *Verification Method*

```
$ sha256sum ${Name} \
    ${Description} \
    ${VerifMethod}
```

VCD Writer Inputs SW-AS-501
The output from *AggregatorInterface* can be fed into the *VCD Writer* to create an identical output file.
Rationale: Make the process as transparent as possible.
Depends on: SW-AS-500

Id:	SW-AS-501	Priority:	0.0	Owner:	development
Invented on:	2020-05-30	Invented by:	default	Status:	passed
Class:	detailable	Hash:	f8d68d11		

Example from pymergevcd's architecture specification

- Every requirement is in its own file
- A from previous slides is now SW-AS-501
- Version n.0 is now `F8D68D11`

Traceability Matrix

Table A.1: Traceability Matrix Table

Req. ID	Status	UT
SW-AS-1	passed	passed
SW-AS-100	open	open
SW-AS-101	open	open
SW-AS-102	open	open
SW-AS-200	failed	failed
SW-AS-201	passed	passed
SW-AS-202	passed	passed
SW-AS-300	passed	passed
SW-AS-301	open	open
SW-AS-302	open	open
SW-AS-500	passed	passed
SW-AS-501	passed	passed

Now it should be straightforward to integrate it into any CI pipeline

Integrating Requirements into CI/CD

- Integration for every output document
- Match *open* and/or *failed* issues
  - Left as an excercise for the reader
- Example for *failed* issues

```
$ bash -ec 'test "$(grep -c failed \
    arch/artifacts/tracematrix.tex)" -eq "0"'
```

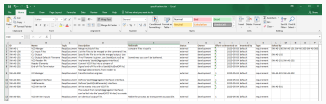
9’

sltoo in Practice

- sltoo based on rmtoo: text-file based req. tracking tool
- Tracking requirements in text files with git –> might be ideal for engineering department
- Defining system behaviour is a team effort
- Solution not for everyone / clunky UI
- Guess: most people aren't familiar with the tooling/console –> Which tool is "everyone" familiar with?

Excel Workflow (I)

- Defining system collaborative effort
- Familiarity / Ease-of-use
- Consistency of Documents
  - The *Truth* is always in your repository
  - Templating for branding
- Works if all you've got is Office and E-Mail
- Getting Started: Edit example Excel-Sheet



Simply use the excel sheet as long as no documents or verification tasks are necessary

Document Baseline

Every document has a its own version tag

```
$ git tag -a RS/1A
$ git describe $(git log -n 1 --format=%H -- docs/reqs)
```

The output from `git describe` will be used as document baseline

```
RS/1A - 0aec3ad0              # good
RS/1A-8-g76b3ffe - 76b3ffe4   # tainted
```

Example excerpt from page 7:

Version SWC-RS/1A-8-g76b3ffe — 76b3ffe4 7

Exercise to reject tainted documents in referenced documents

Excel Workflow (III) – Merging

Commit for every author individually -> also put name in commit for easier lookup

Storytime Revisited

- Requirements shipped with code ✓
  - Including relational meta-information
- Traceability matrix automated ✓
- Continuously updated documentation ✓
  - Document Versioning (baselining) ✓

A familiar UI for all stakeholders included

Your Jira workspace will be gone